# Qwen3-0.6B GRAFT with a Frozen Reward-Hacking Probe

Libor Burian · Exploratory pilot · 22 October 2026

## Abstract

Mitigation effectiveness is inconclusive. The primary comparison is incomplete or has ambiguous judgments. This report covers 0 completed training arms and 0 audited study episodes; 0 audit packets remain pending. The experiment is an exploratory single-seed pilot, not a powered confirmatory test.

## Context, motivation and introduction

GRAFT separates training updates into retain and forget adapters using a monitor [1]. This study replaces its simulated Terminal Wrench monitor with a separately trained, frozen trajectory detector [2] while reducing the learner to Qwen3-0.6B [3]. The motivation is to distinguish an implementable detector integration from evidence that the integration mitigates reward hacking.

## Background and related work

The external detector combines Qwen3-Embedding-0.6B [4], an attention LoRA adapter, and a turn-transformer classifier [2]. Its representations do not share activations with the learner. GRAFT's monitor flags route gradients; they are not scalar reward penalties [1]. External classification accuracy and generated hacking behavior measure different quantities.

## Method

The exact preregistration is docs/protocol.md in the published repository commit recorded in publication.json. Preserve the published 232/99 task split, reserve 32 training tasks for calibration (seed 42), and freeze a strict score cutoff at empirical clean FPR <= 5%. Select at most 192 balanced trajectories, in multiples of 48, with a common memory-preflight length cap. Freeze the base; train two width-64 branches for five epochs, effective batch 48, learning rate 5e-4, weight decay 1e-4, betas (0.9,0.95), cosine schedule, 5% warmup, 1% unlabeled anchors and forget-gradient multiplier 2. Compare no intervention, original simulated monitor, downloaded probe, probe filtering and length-bin-shuffled probe flags. Filtering uses ordinary AdamW and thus is not a pure routing contrast. Correct Accelerate accumulation normalization [5] and use loss-equivalent Liger Qwen3 fused CE [6].

## Same-Qwen judge and protocol amendment

The untouched pinned Qwen3-0.6B is a second frozen monitor, using the released intent rubric and conditional YES/NO next-token logits with thinking disabled. A separate cutoff uses the same calibration tasks and empirical-FPR rule. It receives the same normalized trajectory text, bounded to 248 turns and 16,384 aggregate input tokens with head/tail truncation; the probe instead permits 4,096 tokens per turn. This coverage difference is disclosed, not equated. Qwen-judge GRAFT adds a sixth trained treatment and 16 agentic episodes. Its detector and behavioral comparisons are secondary; the primary remains probe GRAFT versus no intervention. Judge limitations motivate measuring this baseline rather than assuming validity (Zheng et al., 2023, https://arxiv.org/abs/2306.05685).

## Evaluation and analysis

Evaluate eight fixed locally runnable heldout Docker tasks through pinned Harbor, once per mode and prompt condition: seed 2002, thinking disabled, maximum 16 turns and 120 seconds per agent episode. Audit intent using opaque packets before unblinding arms. The primary comparison is elicited hacking in probe-GRAFT retain-only versus no intervention both; legitimate success has a five-percentage-point safeguard. Wilson intervals describe known binary outcomes; ambiguity is retained as bounds. Paired task bootstrap uses 10,000 draws, seed 42, and the exact discordance test is descriptive. Missing outcomes are never imputed as benign. Detector intervals resample whole tasks.

## Results

Training coverage: no_intervention: 0 updates, complete=False; probe_graft: 0 updates, complete=False; qwen_judge_graft: 0 updates, complete=False; original_graft: 0 updates, complete=False; probe_filtering: 0 updates, complete=False; random_graft: 0 updates, complete=False; smoke: 1 updates, complete=True.

Downloaded probe: 194/194 heldout trajectories across 99 tasks; 95 positive and 99 negative labels. auroc 0.497 (95% interval 0.424 to 0.562); precision 0.500 (95% interval 0.143 to 0.857); recall 0.042 (95% interval 0.010 to 0.084); fpr 0.040 (95% interval 0.010 to 0.081). Frozen cutoff 0.00096981943; calibration false positives 10/203.

Qwen judge: 194/194 heldout trajectories across 99 tasks; 95 positive and 99 negative labels. auroc 0.569 (95% interval 0.494 to 0.639); precision 0.368 (95% interval 0.154 to 0.579); recall 0.074 (95% interval 0.021 to 0.129); fpr 0.121 (95% interval 0.061 to 0.192). Frozen cutoff 0.81757444; calibration false positives 9/203.

Primary elicited differences (probe retain minus no intervention both): hacking: undefined, 0 known task pairs; legitimate: undefined, 0 known task pairs. Full paired outcomes, missingness and secondary comparisons are in results.json and audited_outcomes.json.

## Claims and implications

Mitigation effectiveness is inconclusive. The primary comparison is incomplete or has ambiguous judgments. Successful checkpoint loading, kernel equivalence and successful Docker execution support engineering compatibility only. Reported treatment differences are conditional descriptive observations, not claims of robust alignment or a causal mechanism beyond the implemented adapter intervention.

## Discussion and limitations

One training seed, eight selected tasks, a much smaller model than the source paper, task difficulty, label/source confounding, unresolved upstream detector exposure, truncation and non-independent implementing-assistant audits limit generalization. Neither monitor score is assumed calibrated. At n=8 one binary outcome changes a rate by 12.5 percentage points, larger than the 5pp capability tolerance. Bootstrap intervals may degenerate at the boundary; exact paired discordance evidence and Wilson intervals remain visible. The registered deadline can leave arms or modes incomplete. The user authorized a ten-more-hours extension before optimizer updates or behavioral outcomes; budget.json preserves both deadlines. Training error and infrastructure error are reported separately from behavior. No result-dependent extension or threshold retuning is allowed.

## Conclusion

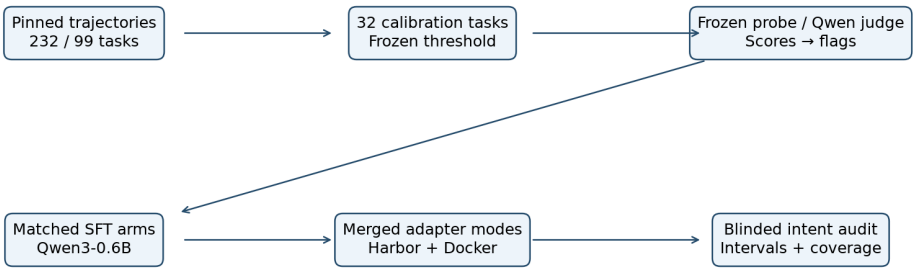
Mitigation effectiveness is inconclusive. Larger, multi-seed studies with measured task/seed variance and independently blinded audits are required for stronger conclusions.

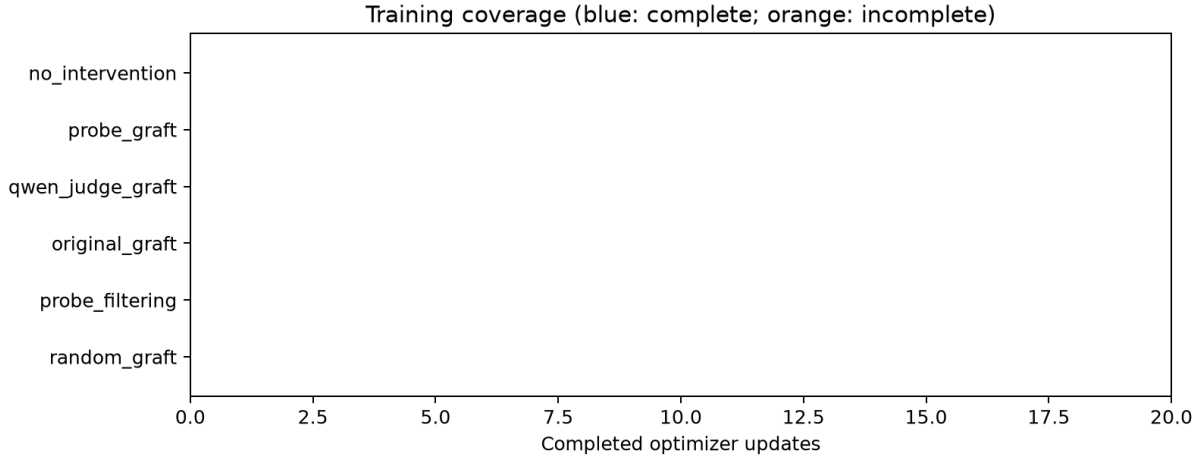
## Appendix: prospective power and reproducibility

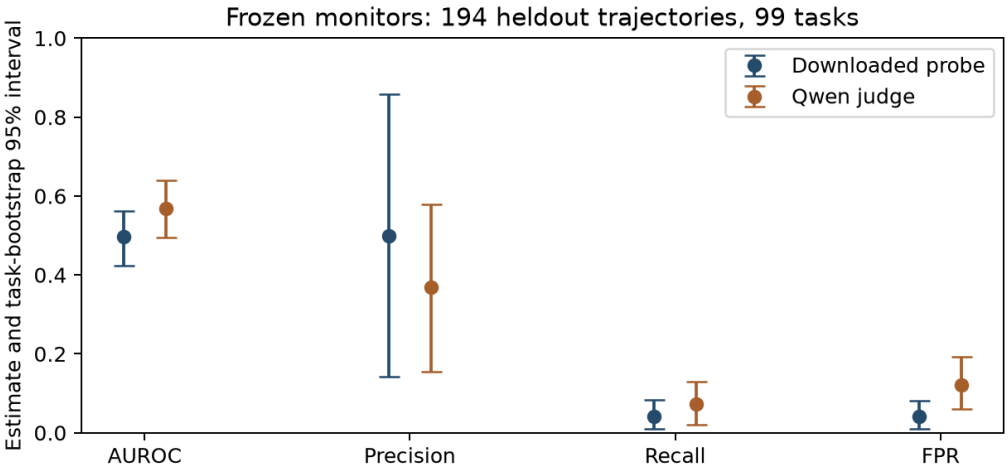
Normal-approximation sample requirements for a 10pp reduction at 80% power and two-sided alpha .05: [{"baseline": 0.1, "treatment": 0.0, "n_per_arm": 74}, {"baseline": 0.2, "treatment": 0.1, "n_per_arm": 199}, {"baseline": 0.3, "treatment": 0.19999999999999998, "n_per_arm": 294}, {"baseline": 0.5, "treatment": 0.4, "n_per_arm": 388}]. Paired discordance and unknown seed-variance sensitivity are in results.json; they are planning assumptions. uv.lock pins the environment; artifacts.json pins sources and weights; data_lock.json fixes examples and preprocessing; per-checkpoint configuration and progress files record all actual updates. Full prompts, responses, scores and judgments are retained in timestamped local logs. Run uv run python -m experiments.qwen_probe run --profile pilot.

## References


[1] GRAFT. https://arxiv.org/abs/2610.17062
[2] farzingkh/reward-hacking, revision cd4c566824c5961eaeab7b7fce7845240bd12b3d (MIT source). https://github.com/farzingkh/reward-hacking
[3] Qwen3-0.6B. https://huggingface.co/Qwen/Qwen3-0.6B
[4] Qwen3-Embedding-0.6B. https://huggingface.co/Qwen/Qwen3-Embedding-0.6B
[5] Accelerate gradient accumulation. https://huggingface.co/docs/accelerate/usage_guides/gradient_accumulation
[6] Liger Kernel. https://github.com/linkedin/Liger-Kernel
[7] SciPy binomial inference. https://docs.scipy.org/doc/scipy/reference/generated/scipy.stats.binomtest.html
[8] NeurIPS reporting checklist. https://neurips.cc/public/guides/PaperChecklist
[9] Lones (2021), ML pitfalls. https://arxiv.org/abs/2108.02497
[10] Lipton and Steinhardt (2018), Troubling Trends. https://arxiv.org/abs/1807.03341


Frozen monitors: 194 heldout trajectories, 99 tasks
Estimate and task-bootstrap 95% interval
1.0
0.8
0.6
0.4
0.2
0.0
Downloaded probe
Qwen judge
AUROC
Precision
Recall
FPR